Market Intelligence Research Brief

Global AI & Data Talent Market — pay, demand and hiring implications

Prepared by: Lisha Bhowmik · **Date:** 2 Nov 2026 · **Built with:** Amazon QuickSuite Market Intelligence Agent (Kaggle dataset + Quick Research) **Rubric map:** 1 Objective & Scope · 2 Approach · 3 Key Insights · 4 Visual Evidence · 5 Confidence & Limitations · 6 Strategic Implications & Next Steps · 7 Leadership Summary · Appendices

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0. Leadership summary (read this first)

AI-skill demand is growing roughly **8× faster than the overall job market**, and pay is structured mainly by **seniority** (executive ≈ 3× entry-level), then **location** (Switzerland, Norway, Denmark and the US lead) and **company size** (large ≈ +28%). Python and SQL are the most requested skills, but **no individual skill shows a reliable pay premium**. Junior pipelines are under pressure.

Decision this supports: how to budget and structure AI/data hiring and pay for 2027. **How far to trust it:** only the demand-growth signal is High confidence. Everything drawn from the Kaggle data is capped at Medium because the data shows synthetic-data signatures; treat dataset figures as *directional benchmarks to validate*, not as market facts. **Recommended actions:** (1) stress-test budgets against rising AI-talent costs; (2) build pay bands on level → location → firm size and validate with a real salary survey; (3) invest in Python/SQL fundamentals and internal training rather than tool-specific premiums; (4) close evidence gaps (dataset provenance, named competitors) within one quarter.

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1. Research objective and scope

Objective.
Determine where pay and demand are concentrated in the global AI/data talent market and what a leadership team planning AI hiring and compensation should conclude.

Research questions.
1. How fast is demand for AI skills growing? 2. What drives pay: seniority, location, company size, skills, work model, education? 3. Which skills are most requested, and do they earn a premium? 4. What is happening to entry-level talent supply?

Scope.

In scope	Out of scope
15,000 AI job postings per file, 2 files, Jan 2024–Apr 2025, 20 countries, 20 titles, 15 industries	Named competitor analysis (dataset has only 16 generic company names)
External signals 2025–26 (PwC, Lightcast, Stanford AI Index, India reports)	Realised compensation (equity, bonus), forecasts beyond 2026

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2. Approach

Step	What was done	QuickSuite component
1. Internal knowledge base	Kaggle dataset (two CSVs, unmodified) uploaded to a Space	Space + agent knowledge source
2. Dataset profiling	Checked structure, duplicates, missing values. **Found the two files share job IDs but are independent samples** (0% matching salaries) → File B = primary, File A = replication check	Agent
3. Internal analysis	Median salary by level, country, title, industry, company size, remote ratio, education; skill frequency; within-level skill premium; monthly postings	Agent
4. External signals	Research objective on demand, pay, skills, work model, entry-level, India; every figure logged with source, date and quality flag (primary / secondary / unverified)	Quick Research
5. Synthesis	Market analysis combining dataset [DS] and external [EXT] evidence	Agent
6. Validation	Each insight re-tested on File A, compared against external sources, rated High / Medium / Low	Agent
7. Leadership brief	This document	Agent

Confidence rule. *High* = replicates in both files **and** corroborated by primary external sources. *Medium* = one of the two. *Low* = neither, or contradicted. Because the data looks synthetic, no dataset-only insight can be High.

3. Key insights

#	Insight	Evidence	Why it matters	Confidence
1	**AI-skill demand is growing ~8× faster than the overall job market.**	[EXT] PwC 2026: jobs requiring AI skills **+69%** vs **+9%** overall (>1 bn ads, 27 countries). Lightcast: AI share of US postings **2.5%**, +55% YoY; agentic-AI postings 0.06% → 0.23%. [DS] Cannot show a trend (postings flat, 842–1,021/month).	Competition for AI talent keeps rising; plans built on 2024 conditions will under-budget.	**High** (direction) / Medium (magnitude)
2	**Seniority is the biggest pay lever.** Median EN $66k → MI $92k → SE $129k → EX $197k (**2.97×**).	[DS] Same in File A (2.94×); correlation of years of experience with salary r = 0.74 in both. Education and benefits score show no effect. [EXT] none retrieved.	Most pay-budget variance sits in level mix, not in title or industry.	**Medium**
3	**A stable high-pay tier: Switzerland, Norway, Denmark, US** pay +44% / +33% / +28% / +15% above the overall median.	[DS] All four rank top 4–5 in both files; 11 of 20 countries agree within 10% across files, but **7 differ by >25%** (India $38k vs $76k; China, Japan, Ireland, Austria, Finland, Israel). [EXT] Lightcast ranks countries by AI-skill *share* (Singapore 4.8%, Hong Kong, Luxembourg, Spain, US 2.6%), which is demand, not pay.	Location strategy for senior hires; the dataset cannot be used for emerging-market benchmarks.	**Medium** (top tier) / **Low** (India, China, others)

#	Insight	Evidence	Why it matters	Confidence
4	**Large companies pay ~28% more than small ones; remote status makes no difference to pay.**	[DS] L vs S: +28% in both files; remote 0/50/100%: $106–109k. [EXT] none for size; remote data (65% of AI postings remote) is unverified and contradicts the dataset's 33% split.	Firm size should be a band factor; do not assume a remote discount.	**Low**
5	**Python (30%) and SQL (23%) are the most-requested skills, but individual skills carry no reliable pay premium.**	[DS] Within-level skill effects −1.4% to +1.7%; ranking does not replicate (ρ = 0.15). [EXT] Lightcast: Python is the #1 specialised AI skill (258,674 US postings, ~+30% YoY). PwC's **62%** AI wage premium is AI vs non-AI work, which the dataset cannot test.	Hire and train for fundamentals; do not pay tool-specific premiums on this evidence.	**Medium** (demand) / **Low** (premium)
6	**Entry-level pipelines are under pressure.**	[EXT] Stanford AI Index 2026: headcount of software developers aged 22–25 fell 2021–25 while older groups grew; 32% of organisations expect AI to reduce headcount. PwC: AI-exposed entry-level roles are 7× more likely to require senior-type skills. [DS] Not testable.	"Buy" strategies for junior talent may fail; internal training may need to fill the gap.	**Medium**

4. Visual evidence

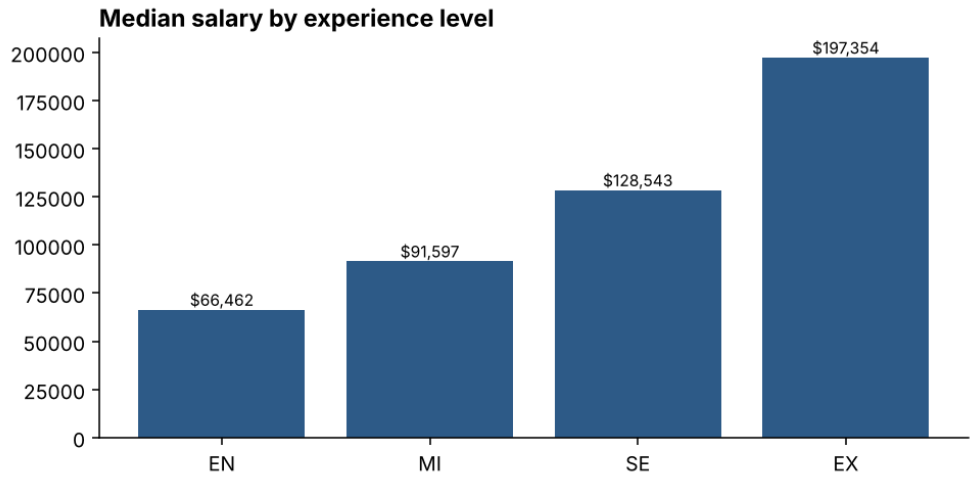


Figure 1: Median salary by experience level (Insight 2)

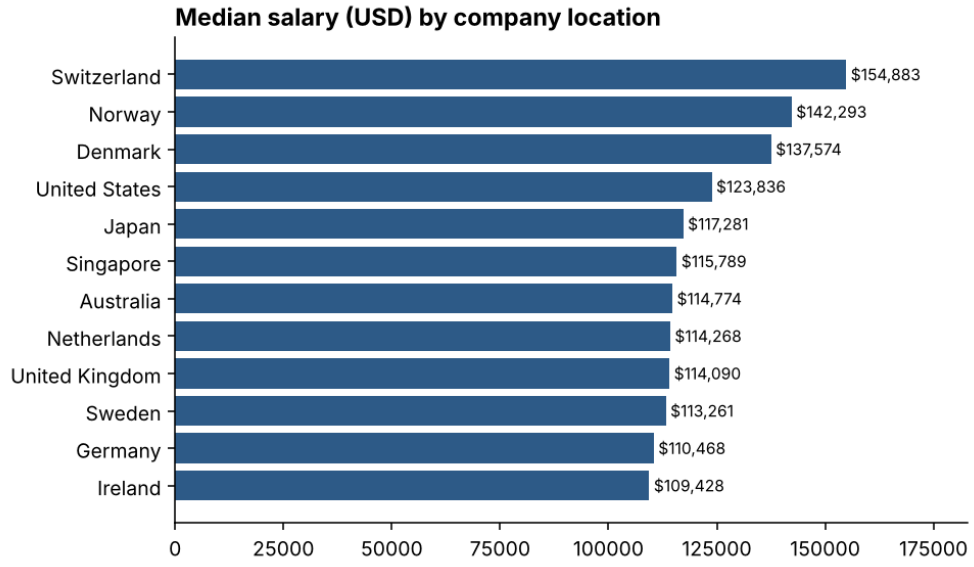


Figure 2: Median salary by company location, File B (Insight 3)

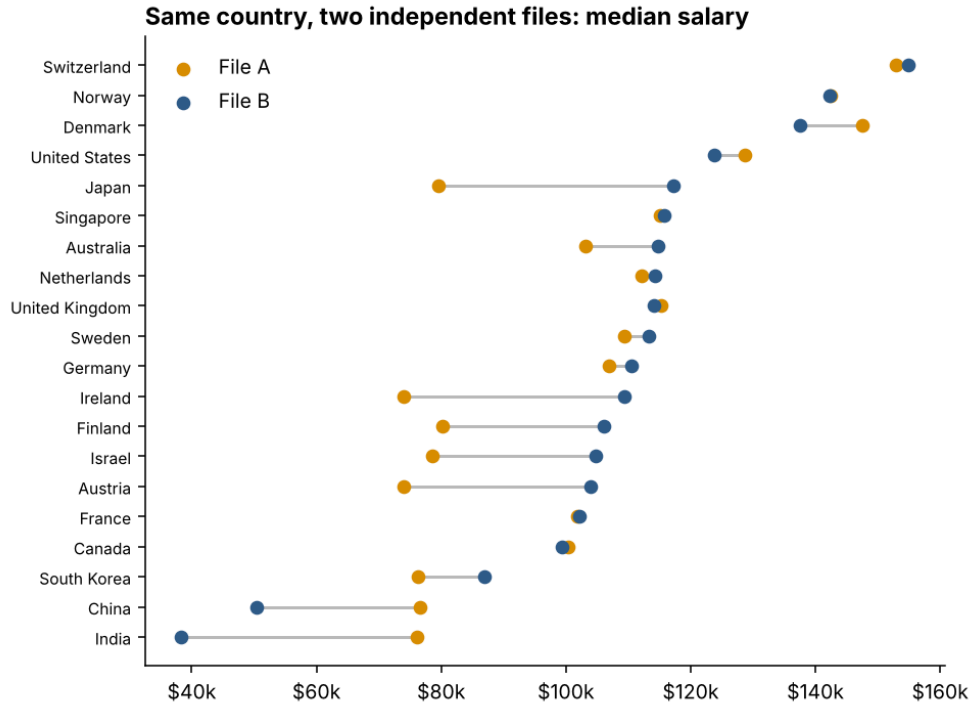


Figure 3: The same country medians in two independent files (Insight 3 and reliability). Top-tier countries match; India, China, Japan, Ireland, Austria, Finland and Israel do not.

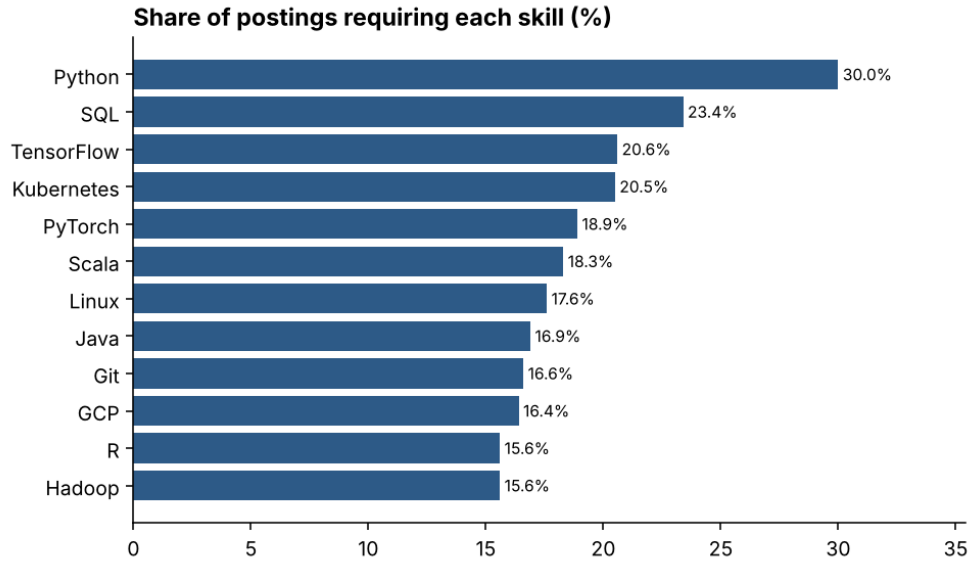


Figure 4: Share of postings requiring each skill (Insight 5)

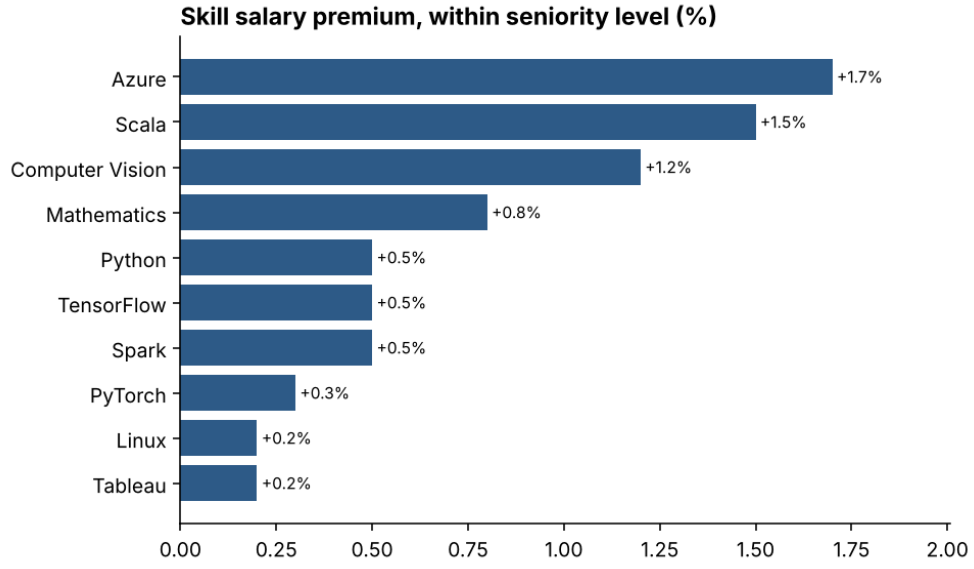


Figure 5: Skill salary premium within seniority level: all effects under ±2% (Insight 5)

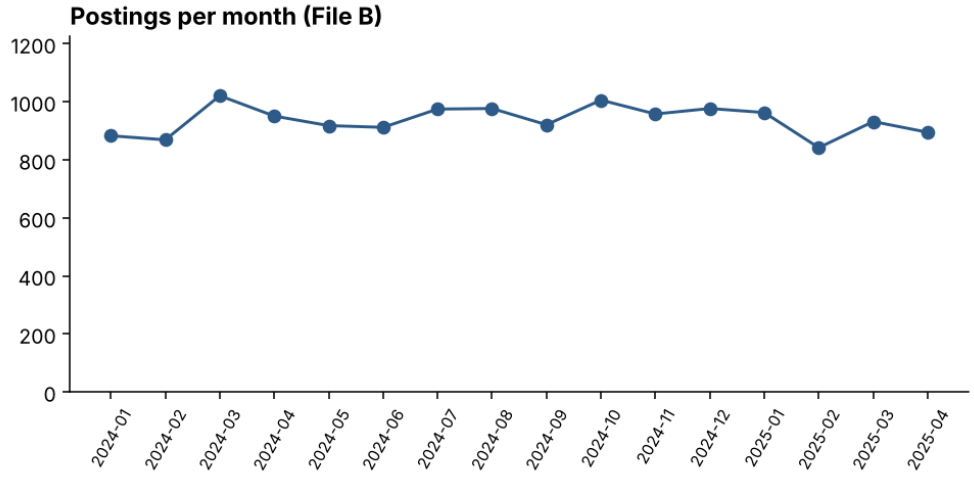


Figure 6: Postings per month are flat, so the dataset cannot show demand growth (Insight 1)

5. Confidence and limitations

5.1 Insight-level evaluation

Insight	Supporting evidence	Gaps / assumptions	Confidence	What cannot be concluded
1 Demand growth	Three independent primary sources (PwC, Lightcast, Stanford) agree on direction	Periods and definitions differ; PwC period not stated on page; Lightcast is US-centric	**High** / Medium magnitude	Growth for a specific country or role; growth after 2026
2 Seniority ladder	Replicates (2.97× vs 2.94×); r = 0.74	No external benchmark retrieved; synthetic risk	**Medium**	Exact multiples for a real employer or country
3 Country tiers	Top 4 replicate; 7 of 20 countries unstable	Cost of living, tax and FX ignored; fixed FX in the data	**Medium** top / **Low** rest	Rankings below the top tier; India and China salaries
4 Size / remote	Size +28% replicates; remote null replicates	Size not controlled for country or seniority; remote conflicts with external claim; no external source for size	**Low**	Causality; real remote-work pay
5 Skills	Python/SQL frequency replicates and matches Lightcast; no-premium result does not replicate	Skill vocabulary is ~25 tags, with no LLM/agentic skills; dataset has no non-AI comparison	**Medium** / **Low**	Premium for any specific skill, in either direction
6 Entry-level	Stanford and PwC (primary)	Stanford page gave no % figure; PwC is US-only	**Medium**	Size of decline; applicability outside the US

5.2 Cross-checks performed

Check	Result
File overlap	Same 15,000 job IDs, 0% matching salary: independent samples
Rank agreement A vs B (country / title / industry / skill premium)	ρ = 0.80 / 0.19 / −0.14 / 0.15: only country (top tier) replicates
Category uniformity (coefficient of variation of counts)	0.02–0.04: implausibly uniform for real postings
Implied FX rates (`salary_local / salary_usd`)	Constant for every row (EUR 0.85, GBP 0.75, JPY 110…): synthetic signature
External sources	PwC, Lightcast, Stanford read directly; Indeed Hiring Lab not readable; Rework (remote share) has no stated method → treated as unverified

5.3 Key limitations

1. **Likely synthetic data.** Agreement between the two files rules out random noise, not a shared generator.
2. **Posting ≠ pay.** Advertised salary only; no equity or bonus.
3. **Timing mismatch.** Dataset ends Apr 2025; external signals run to 2026.
4. **No real employers** → segment-level, not company-level, competitive view.
5. **Raw medians only.** No significance tests or regression controls.
6. **Quick Research gap.** External evidence must be cross-checked against the saved Quick Research output in the Space.

6. Strategic implications and recommended next steps

Implication	Recommended action	Owner / timing
AI-talent costs will keep rising (Insight 1)	Stress-test 2027 hiring budgets with a demand-growth scenario instead of 2024 pay levels	Finance + HR, next planning cycle
Level, location and firm size structure pay (Insights 2–4)	Build pay bands on level → location → firm size; **validate against a real salary survey** (Levels.fyi, Radford, internal HRIS) before adoption	Compensation, within 1 quarter
Fundamentals matter more than tools (Insight 5)	Hire and train for Python/SQL; avoid tool-specific premiums until better data exists	Talent acquisition + L&D
Junior pipelines are thinning (Insight 6)	Pilot internal early-career training pathways to replace external junior hiring	L&D, pilot within 2 quarters
Evidence is not yet decision-grade	Verify dataset provenance; replace with a real posting feed; rerun Quick Research with named competitors and LLM/agentic skill coverage	Analytics, within 1 quarter

7. Leadership summary

The leadership summary is Section 0 on page 1, placed first for quick review.

Appendices (supporting files in the repo)

Appendix	File
A. Dataset reference and profile	`docs/01_dataset_profile.md`, `data/*.csv`
B. Quick Research objective	`research/quick_research_objective.md`
C. External signals log (sources, figures, quality flags)	`research/external_signals.md`
D. Full market analysis	`docs/02_market_analysis.md`
E. Full reliability evaluation	`docs/03_reliability_evaluation.md`
F. Agent configuration and prompts	`agent/agent_config.md`
G. Reproducible analysis code and metrics	`analysis/analyze.py`, `outputs/metrics.json`
H. QuickSuite runbook and checklist	`docs/05_quicksuite_runbook.md`

Sources: PwC 2026 Global AI Jobs Barometer · Lightcast: Four Takeaways from the 2026 Stanford AI Index · Stanford AI Index 2026, Economy · Business Today, 12 Jun 2026 · PropertyCasualty360 on Indeed Hiring Lab · Rework (unverified)